

SURGE PROJECT PROPOSAL

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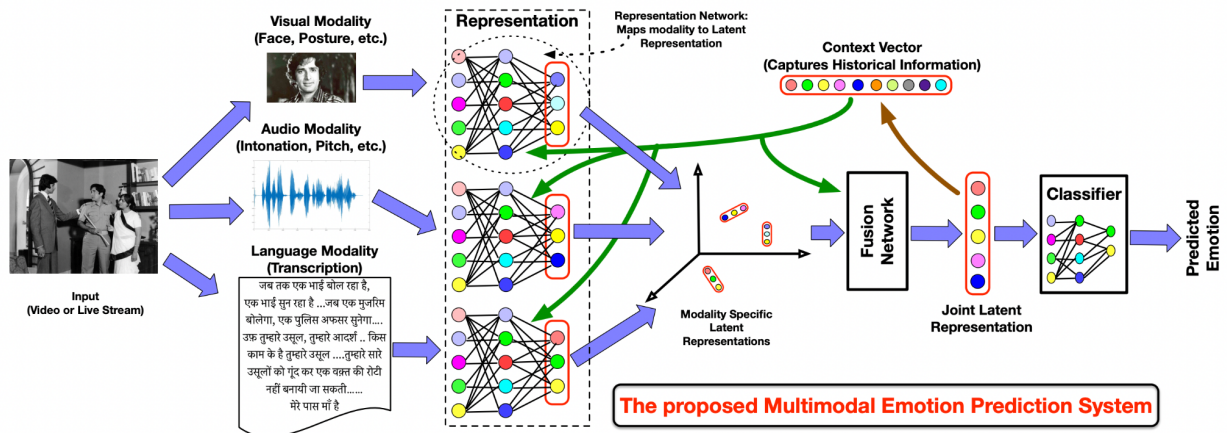
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Project title: Affective computing on Indian Demographics.

Introduction:

The aim of the project is to develop multimodal computational models for the affect (emotion) prediction in the Indian setting. Most of the existing emotion recognition datasets are situated in the Western setting i.e., videos depicting western culture and languages. However, humans express emotions differently in different cultures. A model developed on west-centric corpora would not generalize in the Indian setting. We plan to address this by creating a new multimodal (video with audio and text transcripts) affective corpus to capture the Indian culture and demographics and subsequently, develop deep learning techniques for recognizing affect in the Indian setting (e.g., in Indian videos).

We further plan to study the effect of culture, language, and other demographics on emotion prediction systems. An analysis of change in emotions in different demographics would be helpful to understand the effect of cultural cues present in the corpus.



We plan to use existing and created annotated corpora for developing and training ML models for predicting the emotion of the various people in a given video or live stream. Since humans express thoughts and feelings in different ways (via text, audio, facial expressions, etc.) (Mehrabian and Wiener, 1967; Mehrabian and Ferris, 1967;

Mehrabian et al., 1971; Mehrabian, 2009), we address a multimodal setting, where information from different modalities needs to be taken into account when predicting a person's emotion. We propose developing deep probabilistic models that learn to map signals from different modalities to an emotion label or real value (valence/arousal). Emotion prediction is a two-step process: learning appropriate representations and fusing representations. The figure above gives an overview of the architecture.

Method:

Given the nature and wider scope of the project, it will be done in a group of 3 students. The project consists of two phases

1. Creating Affective Corpus for addressing Indian demographic and cultural setting:

We plan to explore the various available existing movies and TV series with emotion-specific information in this phase. The main objective would be to explore the available TV series/Movies in multiple languages and cultural settings and create a web interface that would facilitate the crowdsourced annotation of the selected corpus.

Further, we aim to collect pilot annotation samples from crowdsourcing and test the developed web interface.

2. Implementing the existing works on Emotion Prediction and comparing the performance on the annotated corpus:

In this phase, we study the performance of existing works on emotions prediction in a multimodal setting. We aim to train existing deep learning architecture for multimodal emotion recognition on the created corpora and compare the performance of various baselines. We further aim to design a new architecture that would generalize to datasets with multiple demographics.

Analysis:

The annotated corpus will be analyzed and compared with the existing corpus to verify the robustness of the created corpora. Further, performance results of the popular existing works would be done by studying the effect of architecture design across multiple datasets.

Expected result:

- ☐ Development of a web interface for annotating India specific TV series / Movies.
- ☐ Creation of Indian multimodal emotion recognition dataset using the developed web interface.
- ☐ Comparison and analysis of the created dataset.

- ☐ Performance comparison of existing neural models for emotion recognition on the created dataset.
- ☐ Development of a new model for emotion understanding via video (face, gestures and body pose), audio, and text modalities.

Weekly Schedule:

Week No.	Dates	Schedule
Week 1	May 16th - May 20th	Exploring various public domain Indian TV series or movies which could be used for creating an Affective computing corpus
Week 2	May 23rd - May 27th	Designing a web interface for annotation of the selected corpus.
Week 3	May 30th - Jun 3rd	Implementation and testing of the created web interface
Week 4	Jun 6th - Jun 10th	Collecting pilot annotations using crowdsourcing
Week 5	Jun 13th - Jun 17th	Exploring existing works on Emotion Recognition in multimodal setting
Week 6	Jun 20th - Jun 24th	Implementing popular baselines for Emotion Recognition System
Week 7	Jun 27th - Jul 1st	Testing the Baseline Performance on the created corpus
Week 8	Jul 4th - Jul 8th	Analysis and comparison of existing works on western and Indian setting